AIR UNIVERSITY / ISLAMABAD, PAKISTAN

# AI-Powered Autonomous Robotic Rehabilitation

A clinician-supervised system for personalized post-stroke upper-limb recovery




| PREPARED BY | PROGRAMME |
|---|---|
| Anas Adeel | CodeAlpha — Robotics & Automation Internship |
| **ASSIGNMENT** | **SUBMISSION DATE** |
| TASK 4 / MINI PROJECT PROPOSAL | 09 November 2026 |

A proposed engineering concept integrating multimodal sensing, movement-quality assessment and bounded assist-as-needed control. This is a design proposal—not a built or clinically validated product.

# 01 / PROJECT OVERVIEW

## Abstract

This proposal presents a clinician-supervised robotic rehabilitation system intended to support upper-limb exercise after stroke. A lightweight arm-support or end-effector robot would combine camera-based pose estimation, inertial measurement units (IMUs), joint encoders and force/torque sensing. A sensor-fusion layer would estimate movement state and confidence; an AI model would assess movement quality, effort and compensatory strategies. A bounded assist-as-needed controller would provide only the support required to complete a clinician-prescribed task. An independent safety supervisor would constrain force, speed, workspace and emergency stopping. A dashboard would allow therapists to configure exercises and review longitudinal measures. The system is proposed for staged prototyping and bench evaluation; no patient testing, clinical validation or recovery benefit is claimed.

## Problem statement

After stroke, upper-limb impairment can restrict reaching, grasping and daily activities. Conventional therapy depends heavily on therapist time and may be difficult to deliver at high repetition while consistently quantifying movement quality. Robotic systems can provide repeatable assistance and measurement, but a fixed assistance profile may not reflect a patient's changing capacity. The proposed research problem is how to combine multimodal sensing and conservative AI assessment to adapt support without transferring clinical authority or safety-critical control to a learned model.

## Motivation and significance

The project explores a practical middle ground between passive automation and unrestricted autonomy: the robot can adjust a narrow assistance parameter within clinician-approved limits, while the therapist defines goals, monitors performance and can stop or reconfigure a session. The intended contribution is an auditable architecture and evaluation protocol for personalized assistance not a claim of superior clinical efficacy.

## Aim and measurable objectives

Aim: design and evaluate a safe prototype architecture for personalized, sensor-informed upper-limb rehabilitation assistance.

- Integrate camera, IMU, encoder and force measurements into a synchronized state estimate.
- Estimate task completion, movement smoothness, range, trunk compensation and interaction effort.
- Implement bounded assist-as-needed control with clinician-set parameters.
- Build a dashboard for exercise setup, session review and exportable progress summaries.
- Verify safety functions and technical performance on a test rig before any human study.

## Intended users and use

Primary users are adult post-stroke patients selected by a rehabilitation clinician; operators are trained physiotherapists or occupational therapists. Initial intended use is supervised, seated, planar or low-speed reaching practice in a rehabilitation laboratory. The system is not intended for unsupervised home use, diagnosis, emergency care or autonomous treatment planning. Eligibility, contraindications and exercise selection remain clinician responsibilities.

# 02 / SYSTEM ARCHITECTURE

The design separates perception, AI assessment, motion control and safety. The learned model produces a bounded assistance request; it does not directly command motor currents or bypass hard limits. The therapist's configuration and the independent supervisor remain authoritative.




*Figure 1. Proposed system data path. The safety supervisor is independent of the AI assessment and can inhibit motion regardless of model output*

## Hardware configuration

| SUBSYSTEM | PROPOSED COMPONENT | FUNCTION / DESIGN NOTE |
|---|---|---|
| Robot interface | Arm support or end-effector; low-inertia actuators | Support arm weight or guide reaching; mechanically adjustable to patient geometry. |
| Visual sensing | RGB/depth camera mounted near workspace | Estimate shoulder, elbow, wrist and trunk landmarks; camera is not sole safety sensor. |
| Wearable sensing | IMUs on upper arm and forearm | Capture segment orientation and angular velocity; calibrate before each session. |
| Robot sensing | Joint encoders and force/torque sensor | Measure robot configuration and patient–robot interaction; detect unexpected loads. |
| Safety hardware | Physical E-stop, limit switches, mechanical stops, watchdog | Independent inhibit path, bounded workspace and controlled release. |
| Computing/UI | Real-time controller + separate AI computer + therapist display | Partition safety-critical control from non-safety AI inference and logging. |

*Table 1. Proposed hardware modules; final actuator sizing and sensor placement require engineering and clinical risk analysis.*

## Software stack

A real-time control process handles encoder/force feedback and low-level motion. A perception service processes camera and IMU streams; a fusion service aligns timestamps and estimates joint/segment state. The AI service calculates movement-quality features and a confidence score. A session manager applies clinician-defined exercise parameters, records events and presents summaries. Communication between services should be authenticated, versioned and monitored for delay or loss.

# 03 / SENSING, AI AND WORKING METHOD

## Session workflow

1. Therapist selects the patient profile, confirms suitability, fits supports and checks sensor placement.
2. The system performs camera/IMU calibration and a low-force range check; the therapist confirms the workspace and target.
3. The patient performs a short baseline set with minimal or no assistance where clinically appropriate.
4. Sensor streams are synchronized, filtered and fused into a movement-state estimate.
5. AI evaluates movement quality and uncertainty; the controller selects a conservative assistance level inside the clinician's limits.
6. The therapist observes the session, pauses or changes the plan when needed, and reviews the session report.

## Sensor fusion and movement assessment

Camera pose estimation provides spatial landmarks and trunk position. IMUs add high-rate segment orientation and angular velocity. Encoders provide robot joint state, while force/torque sensing captures interaction load. A timestamped estimator can combine these signals using an extended Kalman filter or complementary filter, with sensor-health flags and confidence bounds. If a modality is occluded, stale or inconsistent, the estimator should lower confidence and the controller should revert to a predefined conservative mode or stop.




*Clinician-set limits and an independent stop path remain active throughout.*

*Figure 2. Proposed assist-as-needed loop. The robot responds to measured effort and task error, while clinician limits and independent safety checks remain active.*

The AI assessment module would begin with interpretable features: endpoint error, movement time, path efficiency, peak velocity, velocity smoothness, active range, inter-joint coordination, trunk displacement and interaction force. A supervised model could classify movement quality or estimate effort from labelled, consented recordings. Training labels should be defined with rehabilitation specialists, inter-rater agreement measured, and patient-level separation used between training and test data to prevent leakage. The model should return confidence and an "uncertain" state rather than forcing a classification.

AI is advisory to the controller. It may recommend a small change in assistance gain or target difficulty, but only through a constrained interface. No online self-learning is proposed during patient use. Model updates would be versioned, tested offline and released only after review.

# 04 / CONTROL, SAFETY AND EVALUATION

## Assist-as-needed control strategy

The control objective is to support task completion while preserving voluntary effort. A simple initial implementation is impedance or admittance control with therapist-set stiffness, damping, speed and force limits. If the patient initiates movement and remains within the target corridor, assistance is reduced. If progress stalls or the measured effort/trajectory indicates difficulty, the controller may increase assistance gradually within a validated envelope. If pain, excessive force, unexpected resistance, sensor disagreement or loss of communication is detected, motion is halted or safely released. The AI model does not set the envelope and cannot disable the stop logic.

## Safety architecture and risk mitigation

Safety must be designed as a separate, testable subsystem. The independent supervisor monitors position, velocity, force, workspace, sensor freshness, controller heartbeat and emergency-stop state. Mechanical stops and a low-energy design reduce dependence on software. A fault should lead to a defined safe state; the design must avoid trapping the limb or abruptly dropping supported weight. Before use, conduct hazard analysis covering misalignment, excessive torque, unintended motion, entrapment, sensor drift, software faults, electrical hazards, data loss and cybersecurity. Apply ISO 14971 risk-management principles and assess applicability of IEC 80601-2-78 for medical rehabilitation robots, alongside relevant electrical safety, EMC, usability and software lifecycle requirements. [4], [5]

## Evaluation plan and performance metrics

Bench evaluation precedes any human study. Use a compliant test fixture and scripted trajectories to test repeatability, tracking error, latency, force limits, stop distance, fault detection and safe-state transitions. Sensor evaluation should measure pose error against a calibrated reference, synchronization error, drift and dropout recovery. AI evaluation should report sensitivity, specificity, balanced accuracy, calibration and subgroup performance where a labelled dataset exists. Control evaluation should quantify assistance magnitude, task completion, smoothness and stability across simulated impairment profiles.

If bench criteria are met, a future feasibility study would require institutional ethics approval, informed consent, clinical eligibility screening, adverse-event monitoring and prospective registration as appropriate. A supervised pilot should primarily assess usability, safety, protocol adherence and measurement reliability—not claim efficacy. A later controlled trial would compare the system plus usual care with an appropriate dose-matched comparator and measure validated outcomes such as Fugl-Meyer Assessment–Upper Extremity, Action Research Arm Test, activities of daily living, patient-reported experience and follow-up retention. The 2025 meta-analysis found only a small short-term capacity advantage and no clear ADL advantage for robotic versus conventional therapy overall; this proposal therefore treats clinical benefit as an open research question [1].

## Data protection and governance

Use data minimization, role-based access, encryption in transit and at rest, audit logs, defined retention periods and secure deletion. Obtain explicit consent for video capture and secondary model training. Keep patient identity separate from research identifiers. Document intended use, model version, training population, known limitations and update history. A clinician must be able to explain why assistance changed and override it at any time. [2], [3], [6]

# 05 / IMPLEMENTATION AND EXPECTED CONTRIBUTION

## Development phases

| PHASE | DURATION | DELIVERABLE / EXIT CRITERION |
|---|---|---|
| 1. Requirements & risk analysis | Weeks 1–2 | Use specification, clinician interviews, initial hazard log, measurable requirements. |
| 2. Mechanical & sensor prototype | Weeks 3–5 | Bench arm support/end-effector, sensor mounts, calibration routine and emergency stop. |
| 3. Data pipeline & dashboard | Weeks 6–7 | Synchronized sensor capture, visual feedback, session records and therapist configuration. |
| 4. AI assessment prototype | Weeks 8–9 | Offline movement-quality model with patient-level data split, uncertainty and baseline comparison. |
| 5. Control & integration | Weeks 10–11 | Bounded assist-as-needed control; independent safety supervisor; fault-injection tests. |
| 6. Bench verification & report | Week 12 | Test report, limitations, risk controls, reproducible demo and next-stage plan. |

*Table 2. Indicative 12-week student prototype schedule; clinical evaluation is outside this build plan.*

## Feasibility, limitations and expected contribution

The proposed prototype is feasible as a staged mechatronics and software project if access to a safe actuator platform, sensors, a rehabilitation advisor and a controlled test environment is available. Cost and complexity will be driven by the mechanical interface, force sensing, safety-rated components and data collection. The initial system should deliberately limit its workspace and exercise set. Camera pose estimation may fail under occlusion; IMUs require calibration and drift management; force sensors require careful alignment; and a learned movement score may not represent meaningful recovery. The system may also be unsuitable for patients with pain, severe spasticity, contracture, unstable joints or other contraindications identified by a clinician.

Expected contribution is a modular reference architecture, a synchronized multimodal sensing pipeline, an interpretable movement-quality assessment prototype, a bounded assist-as-needed controller and a documented bench-verification protocol. These outputs could support later research, but they do not establish patient benefit or regulatory readiness. A clinical product would require formal design controls, verification and validation, usability engineering, cybersecurity management, quality-system processes and jurisdiction-specific regulatory review.

## Conclusion

This proposal treats autonomy as constrained assistance under clinical supervision. AI estimates movement and informs small adaptations; conventional control and independent safety mechanisms govern physical interaction. The design is intentionally conservative: it prioritizes transparency, measurable technical performance and clinician authority before any claim of therapeutic effectiveness.

## References

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[3] World Health Organization, *Regulatory Considerations on Artificial Intelligence for Health*. Geneva: WHO, 2023. https://www.who.int/publications/i/item/9789240078871.

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[5] International Electrotechnical Commission, *IEC 80601-2-78:2019 Medical Electrical Equipment—Part 2-78: Particular Requirements for Basic Safety and Essential Performance of Medical Robots for Rehabilitation, Assessment, Compensation or Alleviation*, Amendment 1:2024. Geneva: IEC, 2024. https://www.iso.org/standard/83341.html.

[6] U.S. Food and Drug Administration, *Marketing Submission Recommendations for a Predetermined Change Control Plan for Artificial Intelligence-Enabled Device Software Functions*. Guidance for Industry and FDA Staff, Aug. 2025. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/marketing-submission-recommendations-predetermined-change-control-plan-artificial-intelligence.